

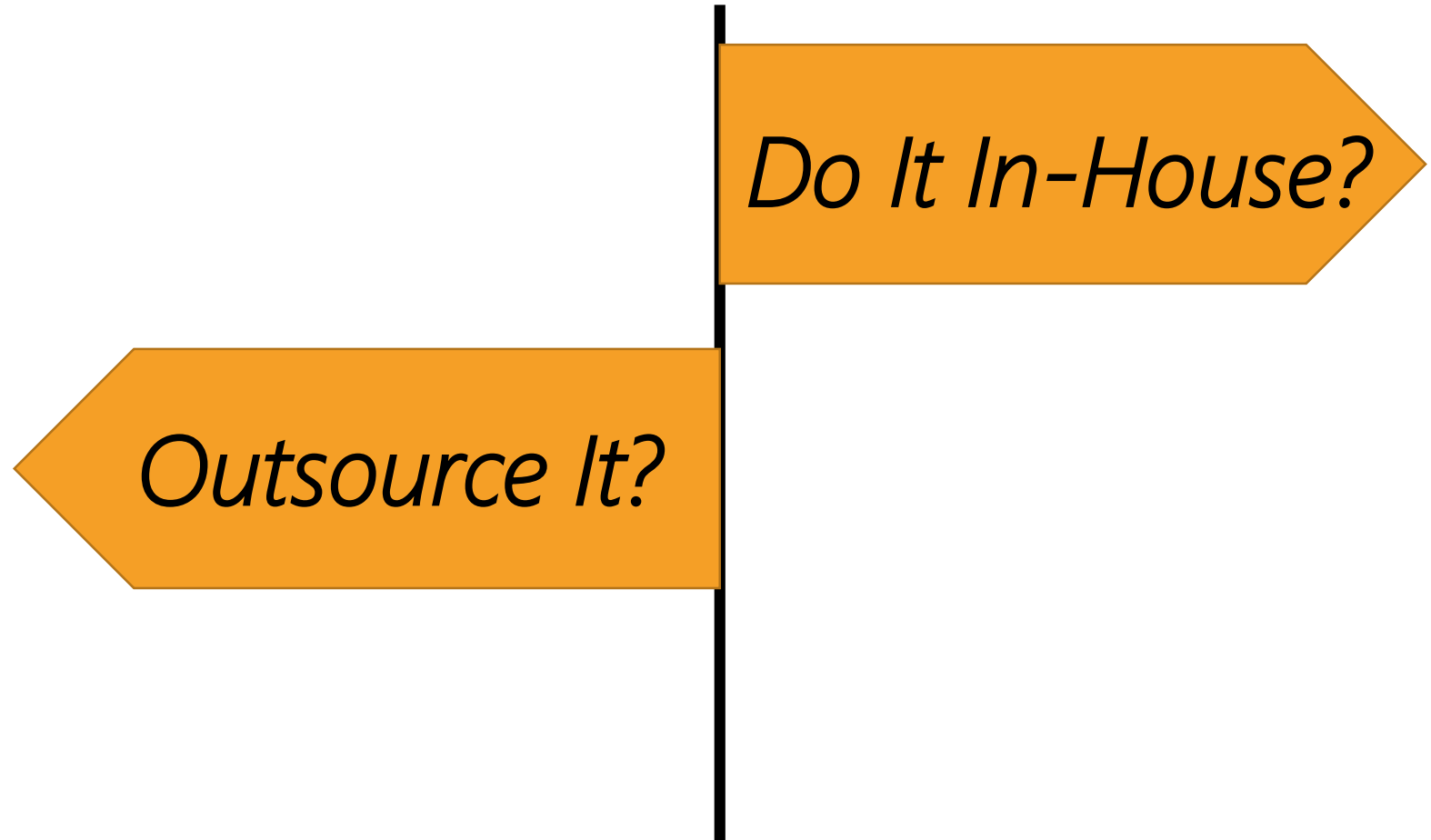


Investment Analysis

• Make or Buy Decision •

A make-or-buy decision is an act of choosing between manufacturing a product in-house or purchasing it from an external supplier. Make-or-buy decisions, like outsourcing decisions, speak to a comparison of the costs and advantages of producing in-house versus buying it elsewhere

• Make or Buy Decision •



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Quality Consideration

If firms that specialize can usually offer higher quality than an organization can attain itself



Cost

Outsourcing may help a firm to avoid fixed costs. Cost savings might come from the item itself or from transportation cost savings



Risk

Liability can be a tremendous risk if the products or services of other companies cause harm to customers or the environment, as well as damage to an organization's reputation. Reputation can also be damaged if the public discovers that a supplier operates with substandard working conditions



Expertise

If a firm lacks the expertise to do a job satisfactorily, buying might be a reasonable alternative



Available Capacity

If an organization lacks the equipment, necessary skills, and time; outsourcing can increase capacity and flexibility



Nature of Demand

When demand is not high and steady, faces wide fluctuations or small orders are usually better handled by specialists who are able to combine orders from multiple sources, which results in higher volume and tends to offset individual buyer fluctuations

• Make or Buy Decision •

Approaches for Make or Buy Decision

- Simple Cost Analysis
- Economic Analysis
- Break-Even Analysis

• Make or Buy Decision •

- ☑ Fixed costs are costs that continue even if no units are produced
 - ☑ Depreciation, taxes, debt, mortgage payments
- ☑ Variable costs are costs that vary with the volume of units produced
 - ☑ Labor, materials, portion of utilities
- ☑ Contribution is the difference between selling price and variable cost

• Make or Buy Decision •

$$\text{Break-Even Unit} = \frac{\text{Fixed Cost}}{\text{SP} - \text{VC}}$$

SP = Selling Price per Unit

VC = Variable Cost per Unit

Break-Even Analysis



• Simple Cost Analysis •

EXAMPLE 13.1 A company has extra capacity that can be used to produce a sophisticated fixture which it has been buying for Rs. 900 each. If the company makes the fixtures, it will incur materials cost of Rs. 300 per unit, labour costs of Rs. 250 per unit, and variable overhead costs of Rs. 100 per unit. The annual fixed cost associated with the unused capacity is Rs. 10,00,000. Demand over the next year is estimated at 5,000 units. Would it be profitable for the company to make the fixtures?

Solution

Cost to make

$$\begin{aligned}\text{Variable cost/unit} &= \text{Material} + \text{labour} + \text{overheads} \\ &= \text{Rs. } 300 + \text{Rs. } 250 + \text{Rs. } 100 \\ &= \text{Rs. } 650\end{aligned}$$

$$\begin{aligned}\text{Total variable cost} &= (5,000 \text{ units}) (\text{Rs. } 650/\text{unit}) \\ &= \text{Rs. } 32,50,000\end{aligned}$$

$$\begin{array}{rcl}\text{with unused capacity} & & + \text{Rs. } 10,00,000 \\ \text{Total cost} & & \underline{= \text{Rs. } 42,50,000}\end{array}$$

Cost to buy

$$\begin{aligned}\text{Purchase cost} &= (5,000 \text{ units}) (\text{Rs. } 900/\text{unit}) \\ &= \text{Rs. } 45,00,000\end{aligned}$$

$$\begin{array}{rcl}\text{Add fixed cost associated} & & \\ \text{with unused capacity} & & + \text{Rs. } 10,00,000 \\ \text{Total cost} & & \underline{= \text{Rs. } 55,00,000}\end{array}$$

The cost of making fixtures is less than the cost of buying fixtures from outside. Therefore, the organization should make the fixtures.

• Break-Even Analysis •

- A manufacturer of TV sells TV cabinet at Rs. 500 each. In case the company makes it within the factory, the fixed and variable costs would be Rs. 4,00,000 and Rs. 300 per cabinet respectively.
- Should the manufacturer make or buy the cabinet if the demand is 1500TV cabinet?

Solution

Selling price/unit (SP) = Rs.500

Variable cost/unit (VC) = Rs.300

Fixed cost (FC) = Rs.4,00,000

$BEP = 4,00,000 / (500 - 300) = 2000 \text{ Units}$

Since the demand (1500 units) is less than the break-even quantity, the company should buy the cabinets for its TV production.

• Make or Buy Decision •

There are three alternatives available to meet the demand of a particular product. They are as follows:

- a) Manufacturing the product by using process A
- b) Manufacturing the product by using process B
- c) Buying the product

Cost elements	Manufacturing the product by using process A	Manufacturing the product by using process B	Buy
FC	5,00,000	6,00,000	
Vc/unit(Rs.)	175	150	
Purchase price/unit(Rs.)			125

The annual demand of the product is 8,000 units. Should the company make the product using process A Or B or buy it?

Solution

Annual cost of process A

$$\begin{aligned} &= FC + VC * \text{Volume} \\ &= 5,00,000 + 175 * 8000 \\ &= 19,00,000 \end{aligned}$$

Annual cost of process B

$$\begin{aligned} &= FC + VC * \text{Volume} \\ &= 6,00,000 + 150 * 8000 \\ &= 18,00,000 \end{aligned}$$

Annual cost of Buy

$$\begin{aligned} &= \text{Purchase price/unit} * \text{Volume} \\ &= 125 * 8000 \\ &= 10,00,000 \end{aligned}$$

Answer: The company should buy the product as the buying price is lower than making it using process A or B.



Thank You